

A methodology to assess the criticality of highway transportation networks

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Abstract Assessing the importance of transportation facilities is an increasingly growing topic of interest to federal and state transportation agencies. In the wake of recent terrorist attacks and recurring manmade and natural disasters, significant steps are needed to improve security at both state and metropolitan level. This paper proposes a heuristic procedure using concepts of complex networks science to assess the importance of highway transportation networks using travel time as the performance measure to assess criticality. We demonstrate the proposed technique both in a theoretical network (Sioux Falls network) and in a built-up network to assess the criticality of the major infrastructures that are used to access Manhattan in an AM peak hour. The results demonstrate the efficacy of the procedure to determine critical links in a transportation network.

Keywords Transportation networks · Critical links · Equilibrium · Disruptions

Introduction

Among the different infrastructure facilities, transportation networks form an important component. Assessing the critical facilities in transportation networks involves examining the importance of roadways including bridges and tunnels that carry traffic.

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Post 9/11, there have been several publications by the National Cooperative Highway Research Program (NCHRP Project 20-07 2002, TCRP REPORT 86/ NCHRP REPORT 525 Volume 12 2006 Department of Homeland Security (National Infrastructure Protection Plan (NIPP)), and AASHTO (Ham and Lockwood 2002) that provide guidelines towards measuring and assessing the importance of infrastructure facilities and developing preventive and protective plans. The decision frameworks are exhaustive in terms of factors considered. However, the decision frameworks are limited since they are based on subjective rating of factors such as ability to provide protection, relative vulnerability to attack, casualty risk, environmental impact, replacement cost, etc., and rankings of contributing factors as opposed to more reliable objective measures. At the same time it is possible to perform more accurate quantitative analysis for few of the factors including environmental impact, replacement cost, and economic impact. There is a need to examine such quantitative measures to better characterize the criticality of different facilities in a transportation network. Such an analysis will enable planners to budget their resources optimally to ensure a resilient transportation network.

This paper proposes a measure and a methodology to assess criticality in transportation infrastructure based on an economic quantitative measure—the impact on total travel time. The travel time experienced by a user depends on his/her route choice decision. This decision in turn depends on the congestion in the network which is a function of the route choice decisions of all other individuals in the network. Therefore, determining each user's travel time requires the resolution of a network game with selfish players trying to optimize their travel times. This problem is referred to as the traffic assignment problem and has been extensively studied (i.e.: see Sheffi (1985) for a complete reference). The importance of a facility may then be assessed by examining the negative effect on travel time if the facility is destroyed or disrupted accounting for changes in user decisions consequent to the disruption. This responds to one of the two dimensions of network reliability given by Bell (2000). The other dimension, connectivity, is also addressed since our approach incorporates the topology of the network into account.

The paper is organized as follows. “Review of methodologies to identify importance of nodes/links” reviews the literature related to modeling and the analytical tools used to address the problem of finding important nodes and links in a network. “Assessing criticality based on equilibrium travel time” presents our proposed methodology. “Examples” presents two application examples. Finally, in “Conclusions” we present our conclusions.

Review of methodologies to identify importance of nodes/links

Disruptions can result from a number of different factors such as component failures, natural disasters (e.g., earthquakes), accidents, intentional disruption (e.g., terrorism or military action). Different techniques have been developed to address this problem in multiple domains: the interdiction problem, the most valuable node (MVN), or the most vital edge (MVE) problem. In this section we review the literature related to these problems.

Interdiction problem

The interdiction problem is defined thus: *an agent attempts to maximize flow through a capacitated network while an interdictor tries to minimize this maximum flow by interdicting (stopping flow on) network arcs using limited resources*. It can be interpreted that the interdictor tries to attack the more important links. The deterministic version of this problem is shown to be NP-complete¹, Wood (1993). Wood (1993) proposes flexible integer programming models to solve the deterministic interdiction problem.

The stochastic interdiction problem has been addressed by Cormican et al. (1998). This is defined as “minimize the expected maximum flow through the network when interdiction successes are binary random variables where an attempted interdiction of arc (i, j) is completely successful with probability p_{ij} and is completely unsuccessful with probability $(1-p_{ij})$.” Independence of interdiction successes is assumed, and only a single interdiction may be attempted on any arc. The problem is formulated as a mixed-integer stochastic program and the solution technique is based on a sequential approximation algorithm. Successful computational results are reported on networks with over 100 nodes, 80 interdictable arcs, and 180 total arcs.

An application of the interdiction problem can be found in Church et al. (2004) which also provides a comprehensive methodology review in the interdiction problem. They focus on the loss of service or supply facilities and not on the loss of capacity of a transport link. Two new spatial optimization models called the r -interdiction median problem (RIM) and the r -interdiction covering problem (RIC) were formulated. Both models identify for a given service/supply system, the set of facilities that, if lost, would affect service delivery the most. They define the r -interdiction median problem and the r -interdiction covering problem. The r -interdiction median problem is defined as “given a set of p different locations of supply find the subset of r facilities, which when removed, yields the highest level of weighted distance”. The r -Interdiction Covering (RIC) problem, instead, of the p different service locations we need to find the subset of r facilities, which when removed, maximizes the resulting drop in coverage. Both problems were formulated as integer programs.

Most Vital Node/Most Vital Edge (MVN/MVE) problem

The *MVN/MVE* is a problem defined in Graph Theory: Given a Graph $G = (V, E)$, the MVN/MVE problem is *to find the node or edge that on its removal results in maximum deterioration of the network performance*. This problem has been proven to be NP-hard² (Bar-Noy et al. 1995). A generic performance measure can be the relative drop of the performance caused by a specific damage to a network.

Latora and Marchiori (2005) propose a method to evaluate the importance of an element of the network by considering the drop in the network's performance caused

¹ NP-complete problem refers to a class of problem that cannot be solved in polynomial time. That is no fast solution has been found for them.

² Informally speaking a problem is NP-Hard if and only if an NP-Complete problem can be reduced into and NP-Hard in polynomial time. This means that the class NP-Hard contains the NP-Complete problems.

by its deactivation. A generic infrastructure is characterized by a variable $O(S)$ that measures its performance. They measure the importance of the damage d by the relative drop in performance. In particular, the *critical damage* is the damage D that minimizes a function O .

Another measure of the performance of a network is the increase of the distance between the origin nodes and sink nodes in a maximum flow graph. In this case, Barton (2005) simplifies the problem through the construction of equivalence classes (partitions) on the set of all possible input graphs. The specific graph G may be transformed through simplified transformations in order to determine its equivalence class. Such simplifications may aid the more efficient determination (rather than the naive brute force approach) of a vital edge of a graph G . Barton (2005) does not provide an algorithm to solve the problem. However, algorithms have been developed for finding the most vital edge in a spanning tree where its removal causes greatest increase in weight of spanning tree of the remaining graph, Shen (1999).

Alternatively, throughput has also been studied in the past as a performance measure. Ratliff et al. (1975) focused on finding the n most vital links in flow networks. The n most vital links of a flow network are defined as those n arcs whose simultaneous removal from the network causes the greatest decrease in the throughput of the remaining system between a specified pair of nodes. These n arcs are shown to be the n largest capacity arcs in a particular “cut”. An algorithm is developed based on the idea of sequentially modifying the network such that the “cuts” eventually result in a reduced network with the smallest capacity.

Applications of *MVN/MVE* problem can be found in Grubescic and Murray (2006), to evaluate the potential impacts of losing critical infrastructure elements that are geographically linked; Qiao et al. (2007) to study the allocation of security resources (budget) to water supply networks as to minimize the network’s resilience; and Modarres and Zarei (2002) to address the problem of planning to minimize earthquake damages.

Specific applications in the assessment of transportation facilities

There are several research papers and technical reports on specific applications of models to assess importance of transportation facilities. The NCHRP REPORT 525 Volume 11 (2006) describes an analytical tool to identify and prioritize state-specific transportation choke points (TCPs) according to their potential economic impact on U.S. commerce. However, the models are simplistic and consider only the increased cost of freight movement associated with the detours, and, increased inventory costs imposed by the relative uncertainty of deliveries through the detour.

A more elaborate model is developed by Matisziw et al. (2007). They employ the p -Cutset Problem (PCUP), a network interdiction model, to evaluate the vulnerability of freight movements in Ohio to disruptions in the interstate system. In particular, they analyze the vulnerability of truck flows within Ohio to disruptions in the interstate system.

The above work considers only freight traffic flow. Several other work focus on both passenger and freight flow together. Ham and Lockwood (2002) identify critical assets in the Nation’s highway transportation network. They define *critical assets* as “those major facilities the loss of which would significantly reduce

interregional mobility over an extended period and thereby damage the national economy and defense mobility”. They identify the critical assets based on the following criteria: Casualty Risk, Economic Disruption, Military Support Function, Emergency Relief Function, National Recognition, and Collateral Damage Exposure. However, the methodology adopted to identify economic loss in particular is based on the additional distance of detour ignoring congestion effects.

In contrast, Scott et al. (2006) present a system-wide approach to identifying critical links and evaluating network performance. The approach considers network flows, link capacity and network topology and is based on a measure—the Network Robustness Index (NRI)—of change in travel-time cost associated with rerouting all traffic in the system should a segment become unusable.

Often the importance of transportation infrastructure is accentuated by special scenarios. A case in point is the importance of certain links and nodes in emergency evacuation scenarios. Murray-Tuite (2003) studied the problem of identifying vulnerable transportation infrastructure under emergency evacuation. The problem is represented as a game played between an evil entity and the traffic management agency (TMA). The evil entity seeks roads with higher disruption index and the TMA routes vehicles trying to avoid the vulnerable links. Unlike other transportation network evacuation models, her formulation also describes household decision making behavior in an emergency evacuation.

More recently, advanced modeling techniques based on stochastic programming and variational inequalities have been developed. For example, Liu and Fan (2007) develop a formulation of the network retrofit problem in stochastic programming framework. The problem goes a step further than identifying critical infrastructure; they prioritize network retrofit strategies based on the importance of facilities and available budgets. Chen et al. (2007) developed a network-based accessibility measure using a combined travel demand model for assessing vulnerability of degradable transportation networks. They formulate the combined travel demand model as a variational inequality problem. The methodology adopted in this study is more comprehensive since it considers individual responses across several dimensions of travel choice simultaneously. However, efficient computation techniques for large scale transportation networks may be unavailable.

Assessing criticality based on equilibrium travel time

The application of the aforementioned methodologies to assess criticality in transportation analysis has to be conducted with caution. This is because the performance of transportation networks is inherently dependent on the congestion effects caused by the interaction between driver behavior and built environments. In the static transportation planning/operations context, the congestion effects can be captured using the user equilibrium model. The methodology proposed in this section assesses the criticality by computing the congestion effects based on user equilibrium with and without the transportation link/node.

In transportation network analysis, Wardrop’s first principle states that every user seeks to minimize his transportation cost which under this perspective is the individual travel time. The flow that satisfies this condition, where no traveler can

improve his/her travel time by unilaterally changing route, is referred as the *user equilibrium* (UE). The problem involves the assignment of origin and destination (O-D) flows to the network links such that the travel time on all used paths for any O-D pair equals the minimum travel time between the O-D. The equivalent mathematical formulation is

$$\text{Min} Z\{T_{ij}\} = \sum_a \int_0^{V_a} C_a(v) dv \quad (1)$$

s.t.

$$\sum_r T_{ijr} = T_{ij} \quad (2)$$

$$V_a = \sum_{ijr} T_{ijr} \delta_{ijr}^a \quad (3)$$

$$T_{ijr} \geq 0 \quad (4)$$

where, T_{ijr} is the number of trips between the O-D pair (i, j) that uses path r , C_a is the cost of flow v using link a , and V_a is the flow in link a , and δ_{ijr}^a equal to one if path r between i and j uses link a and zero otherwise.

Our approach includes the computation of the UE solution, and the assessment of the criticality of the facility in terms of the equilibrium travel time. The transportation literature has developed extensive solution approaches for estimating the UE solution. In this paper, we assume the usual conditions—symmetric cost functions, single user class, inelastic demand (however the measure can be extended to elastic demand models) and perfect information to all users.

Ranking and criticality measure

To define the criticality measure we use the following notations and definitions:

- $G = (N, E)$ Original network, where N is the set of nodes and E the set of edges
- $G' = (N', DE')$ Disrupted network, where N' is the set of remaining nodes and E' the set of remaining edges
- $N^* = N - N'$ Set of nodes to be deleted (disrupted)
- $E^* = E - E'$ Set of edges to be deleted (disrupted)
- DUE^* Deterministic User Equilibrium for G

A generic measure of criticality in a network can be defined by the change in the performance of the network after the removal or damage of one its components. Therefore, the criticality of any of its components can be expressed by

$$D(i) = \frac{\Phi_{G'}(i) - \Phi_G(i)}{\Phi_G(i)} \quad (5)$$

where $\Phi_G(i)$ is the performance measure of the network without disruption and $\Phi_{G^*}(i)$ is the performance measure of the network after the disruption of the component i . The key in using this expression is finding an appropriate performance measure for a transportation network.

One potential measure is the length of the shortest path. Let's define a set of origins and destinations as subsets of N . If there exists a path connecting any O-D pair, the distance d_{ij} between these two nodes is positive and if there exist no path then $d_{ij} = \infty$. The shortest path length l_{ij} between nodes i and j can be defined as the smallest sum of the physical distances throughout all possible paths. Latora and Marchiori (2005) use this measure to assess criticality. However, this measure is not suitable to address effect of congestion in transportation networks.

For a transportation network, an appropriate measure is the *equilibrium* travel time which satisfied user equilibrium (UE) conditions. Under UE conditions, each user's choice is in response to the congestion levels on the network. The travel times obtained at each link captures the underlying behavior of the users in the network. Therefore, we use the aggregated value of travel time over all users as a measure of performance.

The measure is given as the summation of all arc travel times (t) represented by:

$$\Phi_G = \sum_{\forall a} t_a(x_a) \quad (6)$$

where x is the flow at link a . If there is at least one path connecting any O-D pair this value is a positive number but if there is not a path the travel time will become infinite. Hence, we also assume that there is typically path choice between any two given O-D pairs. This measure of criticality differs from Nagurney and Qiang (2007, 2008) definition. They developed a measure that is an average network efficiency matrix that does not count a pair that has no associated demand. In our measure we do not weight our disutility measure by the demands. That is that even if an origin or destination node is disrupted, we do not eliminate the demand associated with it.

Moreover, our performance measure can be extended to address cases when the demand is a function of the level of service of the network (for instance it is conceivable that demand shifts will occur when a new bus service is introduced), i.e., elastic demand. In the elastic demand case, the performance measure is again Eq. 6 but the trip rate between the O-D pairs is not necessarily deterministic but will be influenced by the level of service in the network, see Sheffi (1985). In that case, the performance measure.

Applying Frank-Wolfe algorithm to assess criticality

To present our methodology we first state that our assumption is that, under Wardrop equilibrium each vehicle seeks to minimize its journey time. We have used this principle to assess the criticality of the links by determining the change in total travel time due to the deletion of a link or a node. The algorithm is based on the convex combinations algorithm (for details of the algorithm the reader is referred to the books by Sheffi (1985) or Ortúzar and Willumsen (2006), also called the Frank-Wolfe.

Our proposed algorithm is an iterative process of choosing one link and eliminating or reducing its capacity. In each iteration of the algorithm, the UE

Figure 1 Algorithm pseudo-code to assess criticality

Main (Ranking of nodes and links, given G, N^*, E^*)

```

1   Initialize  $T \leftarrow 0$ 
2   For all  $n^* \in N^*$ 
3        $E^* \leftarrow$  Find entering/exiting links to/from  $n^*$ 
4        $E \leftarrow$  capacity of  $e^* \approx 0$ 
5       For all  $n \in N$ 
6            $T \leftarrow$  Compute  $DUE(G)$ 
7       For all  $e^* \in E^*$ 
8            $E \leftarrow$  capacity of  $e^* \approx 0$ 
9           For all  $e \in E$ 
10               $T \leftarrow$  Compute  $DUE(G)$ 
11              Ranking( $T$ )

Ranking (given  $T, DUE^*$ )
1   For all  $(n^* \in N^*) \cup (e^* \in E^*)$ 
2        $R \leftarrow (DUE^* - T) / DUE^*$ 
3       Sort( $R$ )

```

solution is computed for the disrupted network. This process is repeated until all links have been evaluated. Finally, the algorithm compares the results with UE solution without disruptions. The links are subsequently ranked using the measure defined in “[Ranking and criticality measure](#)” (see Eq. 6). The algorithm pseudocode is presented in Figure 1.

Examples

Network description 1

We test our methodology on a well-tested transportation network, Sioux Falls Network (see Figure 2). The network consists in a total of 76 links and 24 nodes. Nodes 1, 2, 3 and 13 are the origins and nodes 6, 7, 18 and 20 the destinations. The impedance function is the BPR function with the Alpha parameter set at 0.1 and the Beta parameter set at 2. The additional information such as length, number of lanes, speed is shown in Table 1. The convergence rate for the UE algorithm has been set at 0.001.

We run our algorithm to evaluate the importance of each one of links. We set up an experiment that includes the evaluation of the criticality of the links under different levels of demand. Three different OD matrices: Low Demand (LD), Medium Demand (MD) and High Demand (HD) are tested (see Table 2). The values have been chosen arbitrarily and represent the number of trips for each OD pair (in Figure 2 we presented the nodes selected as origins and destinations).

Analysis of results

Table 3 shows the results obtained for the three demand cases. The table includes only the links whose removal have produced an effect in the criticality ratio, includes only the links that have an effect, $\Phi > 0$. For each link, we compute the criticality measure after it is removed and the next column presents the original V/C ratio for the corresponding link in the complete network (without any disruption). Several interesting observations can be made from the results. The first observation is that as

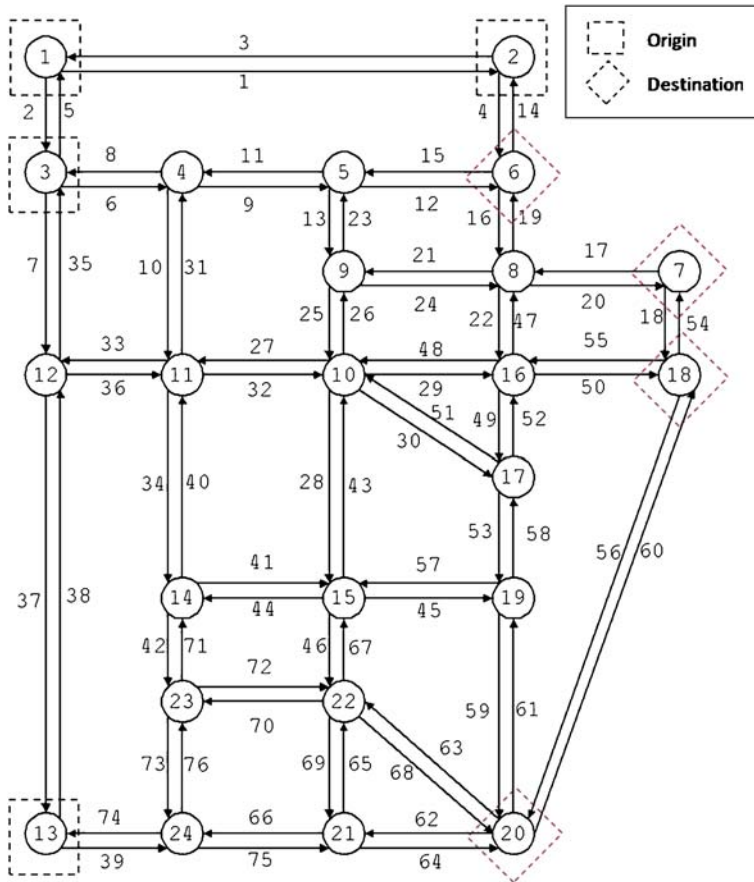


Figure 2 Example 1: Sioux Falls Network

the demand increases the number of links that appear in the ranking increases; there are 21 links in the ranking for the Low Demand case, 26 for the Medium Demand case, and 24 for the Higher Demand Case. The result that the Medium Demand case has 2 more links than the High Demand case appears counterintuitive. However, such counterintuitive results are common (for example, see Braess's paradox in traffic networks when individuals behave selfishly. This further highlights the importance of using user equilibrium based formulation to assess criticality of links on a network. A direct measure suggested by other researchers in the literature (i.e.: Latora and Marchiori 2005) based on shortest path distances will not capture such counterintuitive results. The counterintuitive result notwithstanding, the criticality measures are lower in Medium Demand case compared to those obtained in the High Demand case. If we observe only those values that have a criticality measure of 20% or higher, the number of links affected increases with the demand (see Figure 3). This can also be attributed to the marginal effects of the disruption in congestion. Under Medium demand condition, the link destruction can result in congestion and larger drops in speeds as compared to high demand conditions where speed reduction will be lesser since the network is already congested.

Table 1 Data for Sioux Falls Network.

Link	Length	No. lanes	Cap	Speed limit	Link	Length	No. lanes	Cap.	Speed limit
1	0.4	1	1,800	25	39	0.9	3	2,200	50
2	0.3	3	2,200	50	40	0.3	2	1,800	25
3	1	3	2,200	50	41	0.5	2	1,800	25
4	0.2	2	1,800	25	42	2.9	2	1,800	25
5	3.2	2	1,800	25	43	0.6	1	1,800	25
6	0.2	2	1,800	25	44	0.5	2	1,800	25
7	0.1	2	1,800	25	45	1.2	2	1,800	25
8	0.4	1	1,800	25	46	1.3	2	1,800	25
9	0.4	1	1,800	25	47	1.2	2	1,800	25
10	0.2	2	1,800	25	48	1.6	2	1,800	25
11	1.7	2	1,800	25	49	0.2	3	2,200	50
12	1.7	2	1,800	25	50	0.6	1	1,800	25
13	3.2	2	1,800	25	51	0.6	1	1,800	25
14	1.7	3	2,200	50	52	0.2	3	2,200	50
15	0.3	3	2,200	50	53	1.5	3	2,200	50
16	0.9	1	1,800	25	54	2.9	2	1,800	25
17	0.5	1	1,800	25	55	0.6	1	1,800	25
18	0.2	3	2,200	50	56	0.3	2	1,800	25
19	1.7	3	2,200	50	57	1	3	2,200	50
20	3.2	2	1,800	25	58	0.7	1	1,800	25
21	0.3	2	1,800	25	59	0.9	2	1,800	25
22	1.1	2	1,800	25	60	0.5	2	1,800	25
23	0.3	2	1,800	25	61	2.9	2	1,800	25
24	0.2	2	1,800	25	62	0.9	2	1,800	25
25	0.5	1	1,800	25	63	1.6	2	1,800	25
26	0.5	1	1,800	25	64	0.3	2	1,800	25
27	0.2	2	1,800	25	65	0.5	2	1,800	25
28	3.2	2	1,800	25	66	1.6	2	1,800	25
29	1.3	2	1,800	25	67	0.4	3	2,200	50
30	0.5	3	2,200	50	68	0.7	1	1,800	25
31	0.2	3	2,200	50	69	0.5	1	1,800	25
32	0.6	1	1,800	25	70	0.3	3	2,200	50
33	0.6	1	1,800	25	71	0.7	3	2,200	50
34	0.3	3	2,200	50	72	0.3	2	1,800	25
35	1.1	2	1,800	25	73	1.2	2	1,800	25
36	0.6	1	1,800	25	74	0.3	2	1,800	25
37	0.3	2	1,800	25	75	0.4	2	1,800	25
38	1.9	3	2,200	50	76	0.4	1	1,800	25

Table 2 Origin-destination matrices (origins at nodes 1, 3, 3 and 13 and destinations at 6, 7, 18 and 20).

Origin\Destination	6	7	18	20
Low demand				
1	0	680	550	800
2	500	0	600	625
3	750	600	0	514
13	800	500	700	0
Medium demand				
1	0	1,360	1,100	1,600
2	1,000	0	1,200	1,250
3	1,500	1,200	0	1,028
13	1,600	1,000	1,400	0
High demand				
1	0	2,040	1,650	2,400
2	1,500	0	1,800	1,875
3	2,250	1,800	0	1,542
13	2,400	1,500	2,100	0

The second analysis relates to test the importance of arterials compared to access streets in the network (see Table 4). All links that have a speed of 50mph are arterials and the remaining links with speeds of 25 mph or less are considered access streets. For all cases there are 5 arterials used but as long as demand increases, the criticality measure increases for all arterials. Comparing the LD case and the MD case, the average measure of the MD and the LD cases are similar but the standard deviation is higher for the MD case because few links become severely congested on removing one of the arterials. In streets, the LD has a higher value than the MD case but again, we need to observe the standard deviation and the number of links affected. Intuitively, this means that when the demand increases, users try to use alternative routes through streets; if these streets are disrupted it affects the travel time and our criticality measure captures this behavior. This also highlights the need to consider the entire network and examine the corresponding importance of each facility instead of focusing only on arterials.

Network description 2: New York City

For this test we evaluated the importance of the main access infrastructures to Manhattan Island. The network consists in the four main zones that compose NY City: Bronx, Queens, Brooklyn, and Manhattan. We have also included all New Jersey counties (see Figure 4). The infrastructures considered are the following bridges, tunnels and highways:

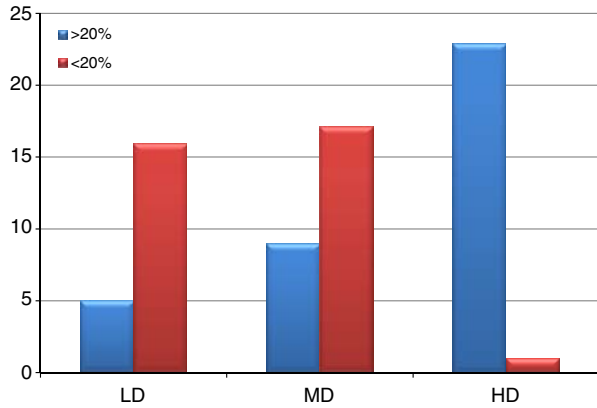
- New Jersey-Manhattan:
 - Lincoln Tunnel (two sections): To Manhattan and to N. Jersey.
 - Holand Tunnel (two sections): East Bound (to Manhattan) and West Bound (to N. Jersey).

Table 3 Critical links in the Sioux Falls Network (includes only the links that have an effect, $\Phi > 0$).

Low demand case			Medium demand case			High demand case		
Link	Critical measure	V/C ratio	Link	Critical measure	V/C ratio	Link	Critical measure	V/C ratio
39	66.17%	9.85%	39	80.34%	19.70%	4	279.50%	41.67%
64	34.25%	18.06%	4	71.78%	28.03%	75	212.65%	46.78%
4	33.81%	14.17%	75	67.87%	32.72%	39	211.51%	31.80%
75	24.63%	18.06%	64	50.02%	32.72%	20	208.21%	17.92%
20	23.32%	6.39%	20	46.23%	12.78%	64	197.63%	46.78%
1	15.64%	15.00%	16	42.96%	24.39%	1	173.86%	42.50%
2	14.56%	3.58%	1	30.43%	28.83%	16	147.83%	35.83%
16	11.89%	12.78%	2	26.19%	7.45%	6	113.04%	26.97%
60	10.23%	6.67%	60	20.17%	13.33%	32	102.15%	36.83%
37	9.82%	6.94%	50	19.09%	28.44%	12	95.69%	8.56%
50	8.89%	14.22%	7	17.61%	13.89%	9	95.53%	17.11%
30	8.89%	3.88%	54	16.08%	7.78%	10	92.10%	18.42%
52	8.89%	3.88%	37	12.65%	13.89%	50	78.15%	36.83%
7	7.66%	6.94%	6	11.02%	20.50%	2	63.10%	11.39%
6	6.51%	9.94%	26	9.02%	1.11%	3	61.37%	4.55%
54	5.45%	3.89%	12	7.84%	5.69%	60	61.33%	24.14%
32	4.74%	14.22%	9	7.79%	11.39%	68	60.15%	23.06%
3	4.25%	1.52%	3	7.71%	3.03%	52	51.33%	10.05%
10	2.85%	7.11%	68	4.63%	6.72%	54	50.58%	12.89%
9	2.25%	5.67%	76	4.61%	6.72%	37	41.29%	24.97%
12	2.25%	2.83%	72	4.48%	3.36%	76	34.17%	23.06%
			30	4.06%	7.76%	72	30.90%	11.53%
			52	3.99%	7.76%	30	26.79%	10.05%
			32	3.84%	29.56%	7	9.75%	24.97%
			10	3.30%	14.78%			
			24	1.40%	0.56%			

- G. Washington Bridge (eight sections): four sections to Manhattan and 4 to N. Jersey.
- Bronx-Manhattan:
 - Croxx Bronx Exp. Bridge (two sections): East Bound and West Bound
 - Macombs Dam Bridge (one section): Both directions Manhattan-Bronx.
- Queens-Manhattan:
 - Queens Midtown Tunnel (two sections) East Bound (to Queens) and West Bound (to Manhattan).
 - Queensboro Bridge (two sections): To Manhattan and to Queens.

Figure 3 Number of links for demand scenario for Sioux Falls Network (LD: Low, MD: Medium, HD: High) by criticality of facility ($\Phi > 20\%$ High Criticality, $0 < \Phi < 20\%$ Low Criticality)



- Brooklyn-Manhattan:

- Brooklyn Bridge (two sections): East Bound (to Brooklyn) and West Bound (to Manhattan)
- Brooklyn Battery Tunnel (one section): Both directions.
- Manhattan Bridge (one section): Both directions.
- Williamsburg Bridge (two sections): East Bound (to Brooklyn) and West Bound (to Manhattan).

Note that these facilities include sections with different directions (to Manhattan or off Manhattan).

The network has been developed in TransCAD using the network included in the New York DOT's Best Practice Model (BPM). The OD matrix considers the 4,000 zones of the NYDOT's BPM for the total AM Peak trips for the year 2002 (about 4.4 million-trips). We coded the algorithm presented in “[Applying Frank-Wolfe algorithm to assess criticality](#)” in TransCAD's language (GISDK) and set up a small toolbox to run the procedures. In the algorithm we measure the criticality as an increase in vehicle hour time (VHT) with the following considerations:

1. The results presented include the final ratio VHT_i / VHT_0 where VHT_i is the total vehicle hours under each scenario and VHT_0 is the total vehicle hours time for the base case (without disruption).

Table 4 Statistical summary for the criticality measure by type of facility (Sioux Falls Network).

Case	Type of facility	Number	Average measure	Std. dev measure
Low demand	Arterials	5	20.55%	12.76%
	Street	16	25.76%	10.64%
Medium demand	Arterials	5	24.46%	29.12%
	Street	21	18.11%	20.86%
High demand	Arterials	5	82.82%	109.71%
	Street	19	73.39%	74.19%

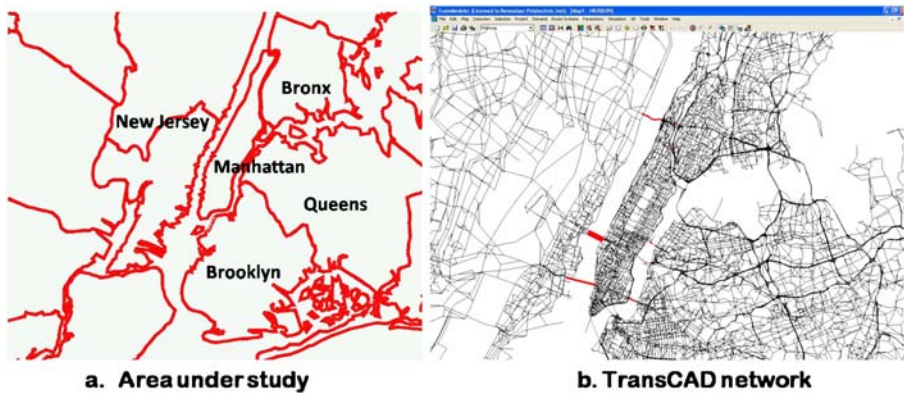


Figure 4 Network example 2: Manhattan main access infrastructures

2. Each scenario is constructed by reducing capacity of each link up to the minimum possible in this case the capacity of each lane was reduced to 1 vehicles/hour.
3. For each traffic assignment, we used 30 iterations for an average convergence rate of 3% in an average of 36 min.

Results and analysis

In this part we will discuss the results of the Manhattan test network. The facilities that are ranked first are the mainly the ones that have direction to and from Manhattan. In the top 10 ranked facilities, only 2 of them carries flow out of Manhattan (Queensboro Bridge and Lincoln tunnel). This is expected since we have used the AM Peak OD Matrix and Manhattan is big attractor of trips. This proves that our criticality measure is consistent with the real pattern of trips and the changes that could possibly occur due to route behavior. We have found that the three facilities that connect New Jersey and Manhattan are the most important. The Lincoln Tunnel and the Holland tunnels section in direction to Manhattan are ranked as the most important links with criticality measures of 52.63% and 39.73% respectively (see Table 5).

An interesting result is that G. Washington Bridge appears ranked 5th. This is due to of the structure of the built line layer. By dividing the George Washington Bridge in eight sections, only one section can be disrupted at a time so the flow using the George Washington Bridge can be split into the remaining sections. Further research can be done in assessing the criticality of all sections of the G. Washington Bridge, but for this paper we are assessing each section independently.

Another interesting result is that the facilities connecting New Jersey and Manhattan are in general ranked first than the ones connecting Brooklyn or Queens to Manhattan. Given that we are using a 4,000 zones OD Matrix that covers the entire region under study (Manhattan, Bronx, Queens and New Jersey) we can infer that this is due to

Table 5 Ranking for the NY City network using the criticality ratio (Φ) and the V/C ratio.

ID	Facility name	Direction	Crit. ratio(Φ)		V/C ratio	
			Value	Ranking	Value	Ranking
90849	Lincoln Tunnel	To Manhattan	52.63	1	5.25	1
90845	Holland Tunnel—EB	To Manhattan	39.73	2	3.15	9
90578	Queensboro Bridge	To Manhattan	14.76	3	1.27	21
160403	Queensboro Bridge	To Queens	10.87	4	1.32	19
90857	G. Washington Bridge	To Manhattan	7.43	5	1.32	8
90861	G. Washington Bridge	To Manhattan	7.43	6	3.40	18
59782	Williamsburg Bridge—WB	To Manhattan	6.59	7	1.36	17
210123	Queens Midtown Tunnel—WB	To Manhattan	6.47	8	2.35	12
90850	Lincoln Tunnel	To New Jersey	5.75	9	2.88	10
90378	Brooklyn Battery Tunnel		5.34	10	0.63	25
90207	Brooklyn Bridge—EB	To Brooklyn	4.79	11	1.16	23
62215	Croxx Bronx Exp. Bridge—EB	To Bronx	4.16	12	3.55	3
58250	Manhattan Bridge		4.13	13	1.20	22
90846	Holland Tunnel—WB	To New Jersey	3.98	14	5.25	2
58212	Brooklyn Bridge—WB	To Manhattan	3.57	15	3.78	7
90318	Williamsburg Bridge—EB	To Brooklyn	2.49	16	2.29	13
61966	Macombs Dam Bridge	Manhattan-Bronx	2.24	17	2.08	15
90854	G. Washington Bridge	To Manhattan	2.03	18	3.90	5
90863	G. Washington Bridge	To Manhattan	2.03	19	2.23	14
90856	G. Washington Bridge	To New Jersey	0.85	20	3.90	6
90862	G. Washington Bridge	To New Jersey	0.85	21	1.67	16
62217	Croxx Bronx Exp. Bridge—WB	To Manhattan	0.78	22	2.76	11
60160	Queens Midtown Tunnel—EB	To Queens	0.77	23	4.26	4
90858	G. Washington Bridge	To New Jersey	0.37	24	0.78	20
90860	G. Washington Bridge	To New Jersey	0.37	25	1.32	24

- That drivers in Brooklyn, Queen and Bronx are not highly affected since they have alternative modes of transportation to travel to Manhattan or they do not require to pass through Manhattan to reach their destinations
- That driver in New Jersey either they mainly have Manhattan as destination or they need to pass through Manhattan to reach their destination.

Another important observation is that only a few critical transportation infrastructures contribute to the greatest loss in travel time (economy) in the region. The disruption of most of the other infrastructures only produces a moderate increase in the increase in travel time. However, the disruption of a few infrastructures produces a large increase in travel time. This can be observed in Figure 4b which presents the map of the region and where only a few links have a thick red line (the criticality of the link in the map is represented by the thickness of the link) and it is summarized in Figure 5. Therefore, our efforts can be focused

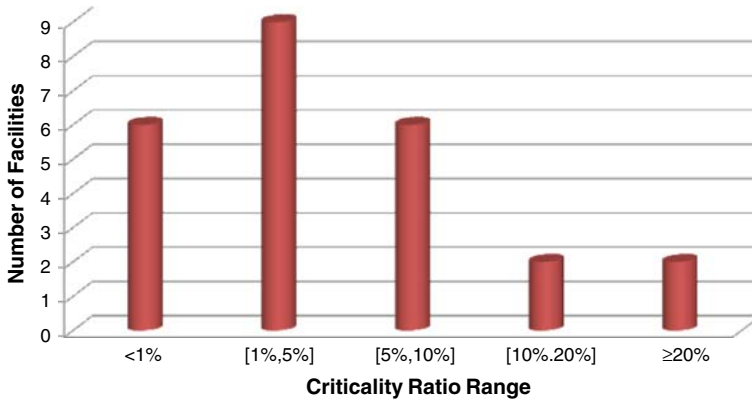


Figure 5 Number of facilities by range of criticality ratio for the Manhattan Network includes only the links that have an effect, $\Phi > 0$

to improve the resilience of these facilities and our criticality measure can be used to identify these facilities.

The last analysis is a comparison of our criticality measure with a traditional measure in transportation, the V/C ratio. When planning the increase in capacity of a link, the V/C ratio is a common measure to define the level of service of the facility. We want to evaluate if this measure also captures the criticality of a link comparing our criticality measure with the V/C ratio in the original network (without disruption), see Table 5. Our intuition can lead us to think that links with larger V/C ratios are the most critical.

In assessing criticality, the situation is different. By observing the values in Table 5, we can note that the V/C ratio cannot capture the effect of route change decision after the disruption. The ranking is different from the one using our criticality measure. For instance the Holland tunnel bound to Manhattan is ranked 9 and the bound to New Jersey is ranked 2nd which contradicts the fact that we are using the AM Peak OD Matrix and we can expect that the bound to Manhattan be more critical.

Conclusions

Knowing the criticality of either a link or a node is important because it could drive our investment decisions in the future. In this paper we presented a revision of current methodologies to assess criticality in a network both by practitioners and scholars. We also provided a practical solution for assessing the criticality of highway transportation infrastructures in a transportation network. Our approach uses travel time as a measure of criticality capturing the congestion effects and can solve for large networks through the use of the Frank-Wolfe algorithm for determining the User Equilibrium solution. The proposed technique is demonstrated on the Sioux Falls network and in a New York City network. For the latter, we assessed Manhattan's main access infrastructures using a network that includes New Jersey, Bronx, Queens, Brooklyn and Manhattan. Our results showed that our criticality measure out-performs the V/C ratio as a criticality measure. This is mainly

because the V/C ratio does not include use behavior which, in some sense, is captured in the traffic assignment. In contrast with the results of our criticality measure, not all the facilities that are going into Manhattan are shown as critical, indicating that just using the V/C ratio is not a good measure to assess criticality in transportation infrastructure. We also found that the disruption of few links represents a higher impact in the travel time making them very critical. Since the AM peak time has been evaluated, we found that all sections that provide the access to Manhattan from New Jersey are more critical. This can be explained by the fact that these two areas have fewer alternatives (in terms of modes of transportation) as compared with Bronx, Brooklyn or Queens.

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